SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I H. Divyaa Dharshini 192426014 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. Divyaa Dharshini

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	9	8	8

Total Marks 25 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

2) CONCEPT MAPPING

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25/30

1) The missing OSI layers are

1) Session layer
2) Data link layer

Functions of the missing layers

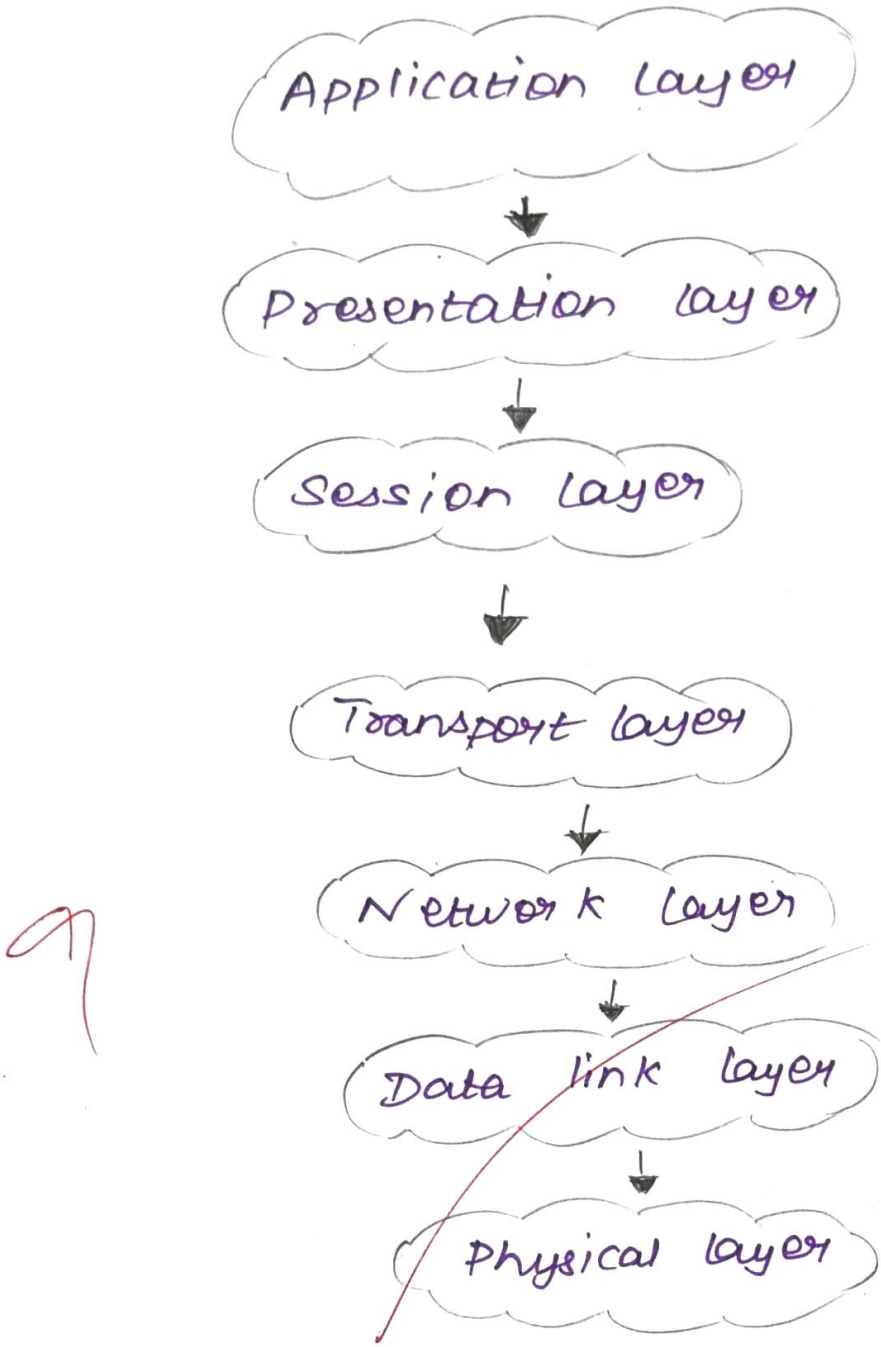
1) session layer:

→ Establishes, manages and terminates communication sessions between devices.
→ Maintains synchronization during data exchange.
→ Helps recover communication after interruptions.

2) Data layer:

→ Transfers data between directly connected devices.
→ Organizes data into frames.
→ Detects transmission errors using error detection techniques.
→ Uses MAC addresses for local delivery.

The session layer maintains organized communi-cation between applications, while the Data link layer helps detect errors during local data transmission. This is how they support reliable communication.


Application layer
Presentation layer
Session layer
Transport layer
Network layer
Data link layer
Physical layer

2)

OSI Layer	Data unit
Transport layer	segment
Network layer	Packet
Data link layer	Frame
Physical layer	Bits.

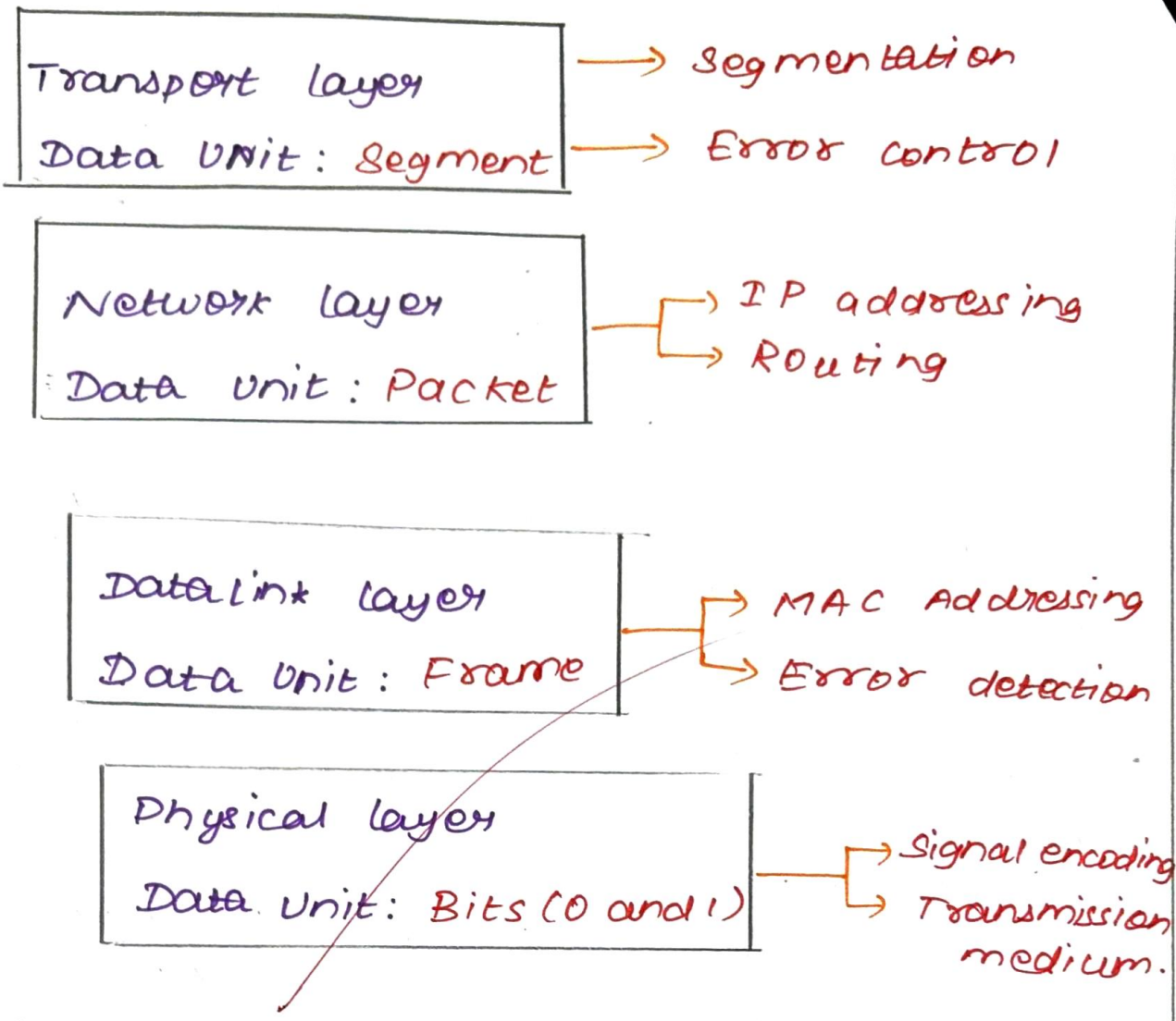
1) segment: The transport layer divides application data into manageable units and adds transport layer information.

2) Packet: The network layer adds IP addressing information to support routing between networks.

3) Frame: The Data link layer adds MAC addresses and error-detection information for local delivery.

4) Bits: The physical layer converts frames into binary bits and transmits them through the physical.

This process is called encapsulation. Each layer adds the necessary control information to the data unit. At the receiving end, the process is reversed through decapsulation allowing the original data to reach the destination application.


Transport layer
Data Unit : Segment
Segmentation
Error control
Network layer
Data Unit : Packet
IP addressing
Routing
Datalink layer
Data Unit : Frame
MAC Addressing
Error detection
Physical layer
Data Unit : Bits (0 and 1)
Signal encoding
Transmission medium.

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Datalink layer - MAC address - Local Delivery

* The Data link layer uses MAC addresses to identify devices on a local network.

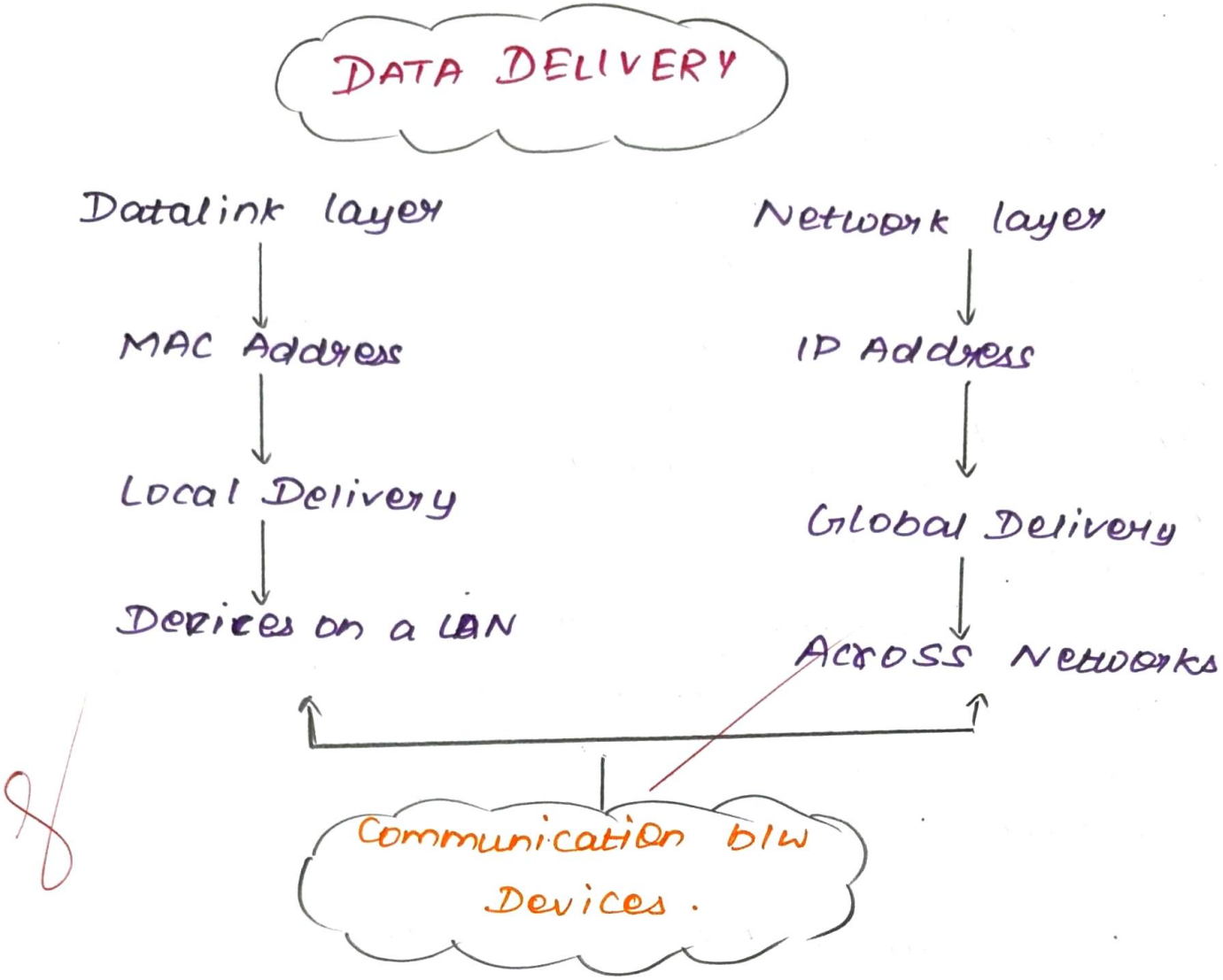
* The source MAC address identifies devices on a local network

* The source MAC address identifies the sending interface while the destination MAC address identifies the next device recieving the frame on the local link.

Network layer - IP Address - Global Delivery

* Network layer uses IP addresses to identify the source and destination across networks.

* Thus, IP addresses supports end to end routing while MAC addresses supports local delivery at each network line.


DATA DELIVERY
Datalink layer
MAC Address
Local Delivery
Devices on a LAN
Network layer
IP Address
Global Delivery
Across Networks
Communication b/w Devices.